

MedTrack_DV

Dashboard Storyboard

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Hospital Operations & Patient Analytics Dashboard Suite

Purpose: Define the layout, analytical intent, filters, navigation, and dashboard interactions before Tableau development. The storyboard follows the project guidance for four interconnected dashboards: Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization.

1. Dashboard Suite Overview

Dashboard	Primary Question	Main Audience	Primary Dataset
Hospital Overview	How is the hospital performing overall?	Administrators / managers	Hospital Overview + Department Analytics + Resource Utilization
Patient Flow	Where are patients moving, and when is demand highest?	Operations / clinical managers	Patient Flow + Hospital Overview
Department Analytics	Which departments are performing well or need attention?	Department managers	Department Analytics + Hospital Overview
Resource Utilization	Are beds, staff, and equipment being used effectively?	Operations / resource managers	Resource Utilization + Department Analytics

2. Global Design & Interaction Rules

- Use a consistent top header with MedTrack_DV branding, dashboard title, and navigation.
- Use a consistent filter area with Date Range, Hospital, and Department. Add context-specific filters where useful.
- Provide navigation between all four dashboards through clearly visible buttons/tabs.
- Use dashboard actions for drill-downs and department/hospital comparisons rather than forcing users to leave the dashboard.
- Keep KPI cards at the top and supporting trend/comparison charts below.
- Use tooltips to show metric definitions, selected values, and relevant operational context.
- Respect dataset grain. Do not blindly join all four datasets because different grains can duplicate measures.

3. Storyboard – Dashboard 1: Hospital Overview

Objective: Provide an executive-level snapshot of admissions, occupancy, patient outcomes, and operational performance.

Area	Proposed Visual / Content	Interaction / Purpose
Header	MedTrack_DV + Hospital Overview title + navigation	Navigate to Patient Flow, Department Analytics, Resource Utilization
Filter bar	Date Range Hospital Department	Global filtering for the selected view
KPI row	Total Admissions Occupancy Rate Avg LOS Readmission Rate Bed Utilization Rate Dept Efficiency Score	Executive summary; values update with filters
Trend row	Admissions trend; Occupancy trend; Readmission trend	Identify monthly/period changes and operational pressure
Patient mix	Admissions by admission type / patient group	Understand volume composition
Department view	Admissions by department + Average LOS by department	Compare workload and stay patterns
Hospital comparison	Hospitals ranked by occupancy or efficiency	Identify stronger/weaker operational performance
Bottom insight area	Admissions vs discharges trend / operational summary	Compare patient inflow and outflow

Wireframe concept



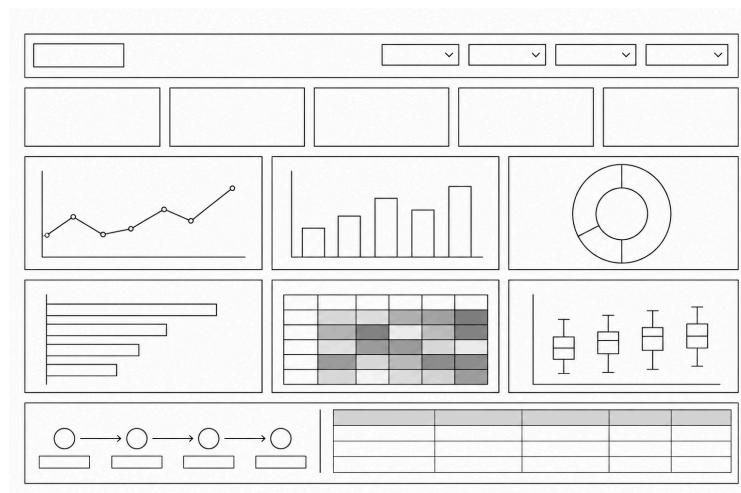
Top	Middle	Bottom
Header + global filters	Admissions Trend Occupancy Trend Readmission Trend	Admissions Mix Department Comparison Hospital Comparison
6 KPI cards	Three analytical trend panels	Operational volume and ranking panels

4. Storyboard – Dashboard 2: Patient Flow

Objective: Analyze admissions, discharges, transfers, department movement, length of stay, and peak patient-load periods.

Area	Proposed Visual / Content	Interaction / Purpose
Filters	Date Range Hospital Department Shift Movement Type	Focus the flow analysis
Flow KPI row	Admissions Discharges Transfers Avg LOS Peak-hour movements	Quick operational status
Movement trend	Movement events over time	Detect high-volume periods
Department flow	From Department → Current Department movement view	Select a department to highlight incoming/outgoing flow
Time analysis	Movements by hour / day of week / shift	Identify peak workload windows
LOS analysis	Average duration in department / LOS distribution	Find departments or periods with longer stays
Patient pathway detail	Admission → transfer(s) → discharge sequence	Drill into selected admission where appropriate

Wireframe concept



5. Storyboard – Dashboard 3: Department Analytics

Objective: Compare department performance, capacity, staffing, patient volume, outcomes, and efficiency.

Area	Proposed Visual / Content	Interaction / Purpose
Filters	Date Range Hospital Department Department Type	Compare selected departments/hospitals
KPI row	Occupancy Rate Admissions Discharges Readmission Rate Mortality Rate Efficiency Score	Department-level performance snapshot
Performance comparison	Department efficiency score ranking	Highlight best/worst performers
Capacity	Total beds vs occupied beds / occupancy trend	Assess treatment capacity
Patient volume	Admissions and discharges by department	Compare workload
Outcome analysis	Readmission and mortality by department	Identify outcome patterns
Staffing	Nurses, doctors, staff-to-patient ratio	Evaluate staffing relative to patient load
Satisfaction	Average satisfaction by department	Add patient-experience perspective

Wireframe concept



6. Storyboard – Dashboard 4: Resource Utilization

Objective: Monitor bed, equipment, and clinical staff utilization, availability, downtime, shortages, and capacity pressure.

Area	Proposed Visual / Content	Interaction / Purpose
Filters	Date Range Hospital Department Resource Type Resource Category	Focus resource analysis
Resource KPI row	Bed Utilization Equipment Utilization Staff Utilization Shortage Count Downtime	Quick capacity and utilization status
Utilization comparison	Utilization rate by resource type	Compare Beds vs Equipment vs Clinical Staff
Bed capacity	Available vs in-use beds	Identify high-capacity-pressure departments
Equipment	Equipment utilization + downtime + maintenance due	Find reliability and maintenance concerns
Staffing	Staff units in use + overtime + response time	Assess staff capacity and workload
Shortage view	Shortage flags by department/resource	Prioritize capacity constraints
Trend	Utilization and downtime over time	Support capacity planning

Wireframe concept



7. Global Filters & Dashboard Actions

Control	Applies To	Behavior
Date Range	All dashboards	Filters time-based metrics; default to the available dataset period
Hospital	All dashboards	Limits the selected hospital and related records
Department	All dashboards	Limits department-level analysis
Department Type	Department Analytics	Compare Emergency, Medical, Surgical, and Critical Care
Shift	Patient Flow	Analyze Day, Evening, and Night movement patterns
Movement Type	Patient Flow	Focus on Admission, Transfer, Department Change, ICU Transfer, or Discharge
Resource Type	Resource Utilization	Switch between Bed, Equipment, and Clinical Staff
Dashboard navigation	All dashboards	Button/tab action between the four views
Department selection	Department Analytics / Resource Utilization	Use a selected department to highlight related charts
Hospital selection	Hospital Overview	Use selection to compare or drill into hospital performance

8. KPI Placement & Definitions

KPI	Formula / Definition	Primary Source
Total Admissions	COUNTD(admission_id)	Hospital Overview
Occupancy Rate	Occupied Beds / Total Beds $\times$ 100	Department Analytics
Average Length of Stay	AVERAGE(length_of_stay_days)	Hospital Overview
Readmission Rate	Readmitted admissions / admissions $\times$ 100; readmission is within 30 days of previous discharge	Hospital Overview
Bed Utilization Rate	Beds in Use / Available Beds $\times$ 100; resource_type = Bed	Resource Utilization
Department Efficiency Score	0–100 composite using occupancy fit, LOS efficiency, readmission, and equipment uptime	Department Analytics